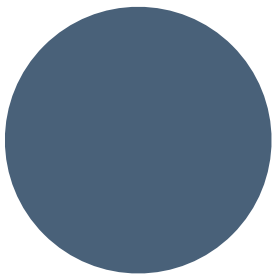


# Grid Resilience Strategy

A long/short equity approach to reward responsible clean energy adoption among US Utilities

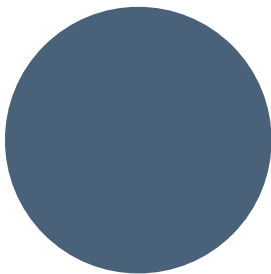


The grid is under stress and not all stocks are equally prepared



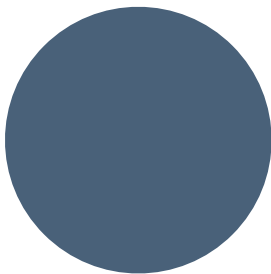
**Grid Fragility is Rising**

Extreme weather events, rising demand from AI data centers, and aging infrastructure are creating increasing stress on the US power grid. Outages and near-misses are growing in frequency.



**Markets Misprice Resilience**

While traditional frameworks emphasize decarbonization metrics and energy prices, we focus on operational resilience — how physical fleet composition, grid position and asset management translate into stability during times of stress



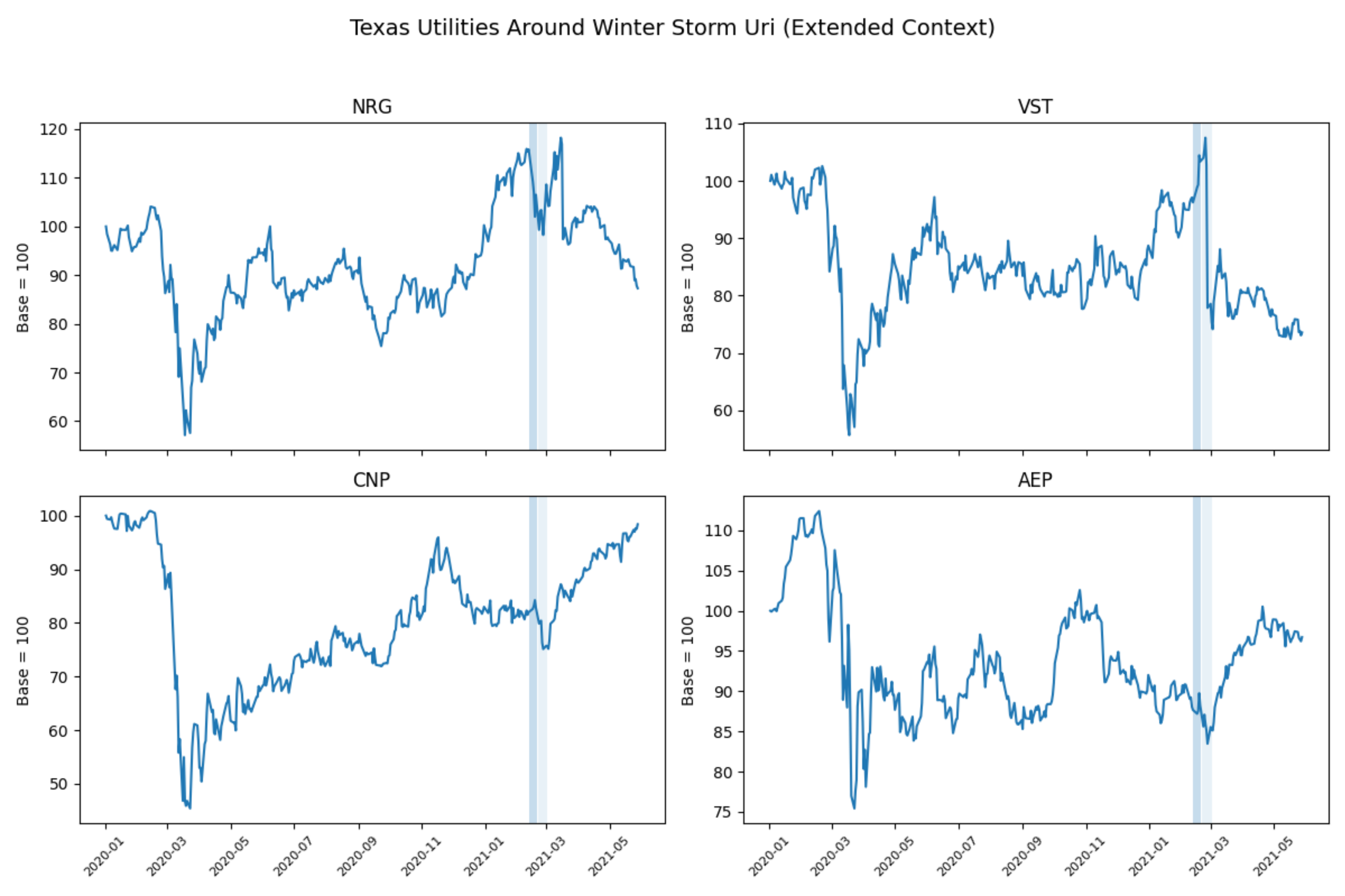
**Responsible Energy Transition**

Our investment approach rewards utilities that have invested responsibly in renewables by rewarding the names that both add renewables to their fleet and also have also responded well during times of known stress/congestion.

**Renewable penetration improves decarbonization — but resilience depends on dispatch ability, storage depth, and grid topology.**

|                | Pros                                                                                                                                                                              | Cons                                                                                                                                                                                              |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supply         | <ul style="list-style-type: none"><li>• No on-site fuel dependency — insulated from gas pipeline or coal delivery disruptions during storms</li></ul>                             | <ul style="list-style-type: none"><li>• Weather-correlated output risk — heavy cloud cover or low-wind profiles can suppress generation precisely when demand is highest</li></ul>                |
| Location       | <ul style="list-style-type: none"><li>• Distributed generation (rooftop solar, behind-the-meter assets) can serve local load when transmission is congested or impaired</li></ul> | <ul style="list-style-type: none"><li>• Utility-scale wind and solar are often remote from load centers. Transmission congestion/ outage can strand renewables during peak stress</li></ul>       |
| Storage Buffer | <ul style="list-style-type: none"><li>• Batteries smooth short-duration spikes independently of real-time renewable output</li></ul>                                              | <ul style="list-style-type: none"><li>• Multi-day events risk storage depletion. Without sufficient recharge capacity, early discharge can reduce system flexibility later in the event</li></ul> |

# NORMALIZED UTILITY STOCK PRICE AROUND WINTER STORM URI 2021



**Stock prices do show reaction around times of grid stress. (Winter Storm Uri)**

Given the richness of grid level data, we can examine high load days (heat domes, winter storms) as well as more subtle but just as damaging transmission events



### Building the Grid Resilience Factor

- **Define the Universe.** Start with a cohesive set of names that are directly exposed to the grid (e.g. Utilities)
- **Neutralize known risk premia.** Remove sector average, style tilts etc
- **Map physical and network exposure:** Use asset ownership to tie utilities to particular generators, load-zones, transmission lines (constraints)
- **Construct Grid Stress Index:** Base this on relative performance at particular nodes. It should reveal systemic structural fragility
- **Estimate Conditional Price Sensitivity:** Measure each utility's (relative performance during times of stress
- **Plays.** Capture stress-conditioned dispersion through sector-neutral spreads —overweight resilient utilities, underweight stress-sensitive peers

The strategy changes behavior precisely when it is profitable

Strategy  
outperforms

As our strategy outperforms, more investment boosts equity prices of top-ranked utilities

Cost of capital falls

Weighted Average Cost of Capital is lowered when share prices rise

Management  
Responds

Boards see valuation premium rewarding responsible decarbonization efforts more than no efforts and more than name-only efforts

Peers Imitate

Noticing the impacts on peers relative to their share prices, boards of our short-position utilities adapt